

Bibliographic Record

Numerical simulation and visualization of flow around rectangular bluff bodies

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Problem Adresses

The main goal of the article is the simulation of unsteady flow around a rectangular cylinder with varying simulation and shape parameters along with different mathematical models for the flow.

By doing this, one hopes to determine to which extent a CFD solution is able to correctly approximate the flow around a bluff body.

Key Ideas

After setting the spacial domain, the article proceeds to initiate different simulations for the flow around the cylinder.

Different meshes are created using different values of Δx for the spatial discretization, from coarse to "finest" mesh. Then it is stated that the flow will be simulated by using different approaches, i.e. LES, RSM and RANS, and, for the latter one, different turbulence models. Moreover, the aspect ratio $\frac{B}{D}$ is modified throughout the simulations. In order to get a turbulent flow, Reynolds number is sufficiently high and fixed: $Re = 21400$.

Methods

Different methods are implemented. Starting from the general approach, we have LES and RSM simulations, along with RANS simulations.

Concerning the numeric schemes, second order implicit method is used for time discretization, while second order upwind method is used for spatial discretization. SIMPLE algorithm is used to solve the discretized equations.

Regardless of the different values of Δx , the domain is always meshed using a hybrid mesh that can account for the different phenomena that happen on a bluff body for this kind of flow. For the RANS simulations, different models for the turbulence are adopted, in particular:

- $k - \epsilon$ model with its standard, RNG and realizable¹ versions
- $k - \omega$ with its standard and SST versions

Applications and Results

Among all the meshes implemented, the article shows how a coarse mesh is not sufficient to obtain optimal results; a fine mesh works better. The flow developed near the obstacle is very complex, characterized by multiple, unsteady vortices. The results also indicates that the Strouhal number is not sensitive to the model used for implementing the turbulence. The LES model is the one which works best but, at the same time, it requires the highest computational cost. Overall, the SST $k - \omega$ is the best choice among RANS models, since it is accurate enough without taking too much computation time.

Errors

Different grids are implemented: the coarse one is not sufficient for a good simulation, because it overestimates the drag force and underestimates other important parameters like the Strouhal number. For this reason, a finer mesh is kept.

The standard $k - \epsilon$ model cannot correctly predict the complex unsteady flow around the cylinder: it underestimates the drag and lift coefficients since it overestimates the energy production in the frontal zone of the cylinder.

Limitations

The simulations have a few limitations. The shape of the body is always fixed and only the aspect ratio $\frac{B}{D}$ changes. It would be interesting considering these types of problems applied to a cylinder whose edges are progressively rounded. The Mach number is kept very low, meaning the fluid is considered incompressible. Finally, the simulations only study a flow with an incident attack angle of exactly zero.

Open issues

First, stochastic pressure fluctuations need a more accurate modeling. Currently, the simulated values are always underestimated compared to the analytical ones. Second, the wake behind the cylinder is naturally irregular and three-dimensional. For this reason, 2D models are not enough to see the true wake details. Full 3D simulations are needed to study these complex flow structures correctly.

¹In this model, some physical constraints are imposed in addition to the standard model.